

FLCR-17 - Continuum Edge Residuals

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper formalizes residual behavior at discrete-continuous edges. The operative object is continuum-edge residual. The core result is that edge residuals can be tracked as finite bridge artifacts without asserting a global continuum limit. The paper also defines how this result routes forward: later GR, condensed-matter, and calibration papers may consume the edge rows with explicit error budgets. Its main residue is explicit: global continuum collapse and unbounded McKay/O2-prime closure remain separated residues.

Keywords

continuum-edge residual; LCR; receipt; claim lane; normal form.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **Continuum Edge Residuals**. The paper's immediate contribution is: Treats continuum edge behavior as sampled residual and numerical-analysis burden.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. Continuum Edge Residuals.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Defines continuum-edge residual as a first-class FLCR object.
- States the local result: edge residuals can be tracked as finite bridge artifacts without asserting a global continuum limit.
- Routes downstream use through claim lanes rather than inherited prose: later GR, condensed-matter, and calibration papers may consume the edge rows with explicit error budgets.
- Preserves the residue boundary: global continuum collapse and unbounded McKay/O2-prime closure remain separated residues.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

2. Source Routing

Primary routed sources:

- 16_Continuum_Edge_Residuals.md
- virtuous/16_Continuum_Edge_Residuals_VIRTUOUS.md
- supplements/16_INTERNAL_REASONING_SUPPLEMENT.md

Cross-corpus evidence classes:

- CAM_CLAIM_100_SCORE_AUDIT.md
- NSIT_TOOL_CLOSURE_MATRIX.md
- NSIT_REASONED_CLOSURE_PASS.md
- PAPER_REWORKS_CRYSTAL_PROJECTION.json
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Continuum edge.** A boundary where discrete receipt data is presented or sampled as continuous data.
- **Residual row.** The named residue left after bridge admission.
- **Receipt boundary.** The exact scope in which the paper's result can be replayed or consumed.
- **Promotion route.** The evidence path required before a stronger downstream claim can use this result.

4. Formal Claims

Claim	Lane	Statement
Theorem 17.1	normal_form_result	edge residuals can be tracked as finite bridge artifacts without asserting a global continuum limit
Proposition 17.2	normal_form_result	continuum-edge residual can be imported by later papers only through its declared source, receipt, and lane.
Protocol 17.3	falsifier_or_open_obligation	global continuum collapse and unbounded McKay/O2-prime closure remain separated residues

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 17.1	formal construction	edge residuals can be tracked as finite bridge artifacts without asserting a global continuum limit	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Proposition 17.2	formal construction	continuum-edge residual can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 17.3	obligation/falsifier	global continuum collapse and unbounded McKay/O2-prime closure remain separated residues	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-17 defines or uses Continuum Edge Residuals as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Identify the local object and its assigned sources for FLCR-17.

- 2. Separate what is internally receipt-bound from what is citation-bound, CAM-bound, calibration-bound, or still an obligation.
- 3. State the strongest admissible claim in its current lane.
- 4. Export only the lane-safe result to downstream papers and preserve the residue.

Proof Sketch

The proof strategy for FLCR-17 is a typed construction argument. The paper names continuum edge as the active object, binds it to the routed legacy and orbital sources, and then allows downstream use only through the declared claim lane. This does not erase stronger ambitions; it keeps them addressable as dependencies, calibration tasks, or falsifiers.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is Continuum Edge Residuals. Its safe consumption rule is:

```
source object -> declared operation -> receipt/lane -> downstream import
```

The unsafe consumption rule is:

```
source phrase -> downstream identity claim
```

That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

7. Evidence And Receipts

Evidence source	What it contributes
Legacy paper body	Primary formal and source-integration material assigned by the series map.
Internal and pyramid supplements	Cross-paper reasoning windows, deferred back-application slots, and route constraints.
NSIT tool closure matrix	Existing verifier/tool families that can close internal or batch claims.
CAM/crystal and standards-window evidence	Projected memory, source routing, standards reports, and window/node databases.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	receipt_bound_internal_result	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.513748.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
causal ledger	normal_form_result	The causal/obligation ledger is closed as a dependency-preserving channel, not as an unresolved label.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-2 tower	receipt_bound_internal_result	The decade tower is resolved with avg TarPit mass=0.510236 and mass sum=40.818854.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-07 cross-lift comparison	cam_crystal_reapplication_result	Family tower binds FLCR-07, FLCR-17, FLCR-27, FLCR-37; avg TarPit mass=0.507568; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
continuum edge residual	normal_form_result	Continuum edge residuals are closed as boundary terms.	Physical continuum limit remains calibration-bound.

Claim Posture After This Pass

FLCR-17 should now be read as a resolved-state contribution for Continuum Edge Residuals at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- FLCR-11 through FLCR-18 for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.
- FLCR-28 through FLCR-30 for CAM/crystal, corpus ribbon, and universal normal-form intake.
- FLCR-31 through FLCR-40 only after the LCR-native result has been cited and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- global continuum collapse and unbounded McKay/O2-prime closure remain separated residues
- External calibration claims require units, datasets, citations, and reproducible data binding.
- A later translation paper may strengthen this result only by adding the missing lane evidence.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): 16/ledger_action.

Original slot	Family	Lift	Role	Proof form
16	ledger_action	1	order-2 slot-6: bind state into causal, observer, or synchronization ledgers	ledger, graph, and synchronization proof

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 15 and reopened at original slot 16 (Continuum Edge Residuals).
It must produce: formal result of original slot 16 plus the same-family lift path toward slot 26.

Semantic transition: bind state changes into causal, observer, synchronization, or obligation ledgers.

Accepted formalism targets to bind in the journal rewrite:

- directed graphs and partial orders
- causal/event ledgers
- synchronization protocols
- traceability and provenance invariants

Slot	Family	Lift	Produced proof form
16	ledger_action	1	ledger, graph, and synchronization proof

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work to turn obligations into graph rows: event, observer, dependency, validator, falsifier, and next lane.

Primary Evidence Inputs

Source	Placement reason
16_Continuum_Edge_Residuals.md	primary assigned source for the paper's formal spine
supplements/16_INTERNAL_REASONING_SUPPLEMENT.md	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations
virtuous\16_Continuum_Edge_Residuals_VIRTUOUS.md	prior enriched paper body; should be mined for mature claims and proof ordering

Secondary And Orbital Evidence Inputs

Source	Placement reason
supplements/SM_BRIDGE_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/RECEIPT_VERIFIER_CATALOG.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_PYRAMID_INDEX.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
CAM_CLAIM_100_SCORE_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_TOOL_CLOSURE_MATRIX.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_REASONED_CLOSURE_PASS.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
ORBITAL_DATA_CROSS_POPULATION_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
AGENT_CRYSTAL_WORK_HARVEST.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body

Source	Placement reason
PAPER_REWORKS_CRYSTAL_PROJECTION.json	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

16	79	partial/system-closeable	continuum edge residuals, Paper 07 bridge, Paper 02 correction, power-of-ten windows	global continuum limit and McKay/O2-prime collapse
----	----	--------------------------	--	--

Paper-Specific NSIT Closure Rows

No direct NSIT row matched the assigned legacy papers in the first-pass matrix; validator matching by object name remains a receipt-attachment requirement.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Lane	Promote now	Bind next	Residue
Symbolic Carrier Versus Physical Carrier	external_calibration_result	Promote addressable symbolic-carrier and transport claims where the paper names carrier, path, residue, or traversal rules.	Map every physical carrier phrase to a standard physics object, Hamiltonian/model, dataset, or calibration route before treating it as measured physics.	Physical identity, units, material constants, and measured transport remain calibration-bound.
Electron-Hole-Exciton Standard Accounting	standard_theorem_citation_bound_result	Treat hole, vacancy, exciton, recombination, screening, band-gap, and effective-mass language as standard-model correspondence when a material model is present.	Attach standard equations/citations and state what LCR adds: addressability, obligation routing, and residue bookkeeping.	Actual material behavior and quantitative exciton claims need a material model, units, and data.
Continuum Edge And Sampling-Stability Bridge	external_calibration_result	Promote discrete-continuous bridge claims only when the sample rule, norm, error bound, or counterexample policy is stated.	Attach sampling rule, interpolation/limit statement, norm, error bound, units, and boundary condition.	Loose continuum, GR, plasma, or energy language stays presentation-level until calibrated.

Binding Discipline

Each queue item is represented as a delta:

local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue

This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Symbolic Vs Physical Carrier	This paper may use carrier language for addressable symbolic transport inside the LCR/CAM system.	Standard physics carrier terminology, local verifier receipts, material/plasma/transport supplements, and calibration routes.	Split symbolic carrier closure from measured physical carrier identity.	Physical identity, units, Hamiltonians, datasets, and measured constants remain external-calibration claims.
Electron Hole Exciton Accounting	Vacancy, complement, hole, residue, and excitation language may appear as LCR addressability/residue vocabulary.	Standard electron-hole-exciton formalism, material-model rows, band-gap/screening/recombinat ion equations, and NP-12 routing.	Treat hole/exciton vocabulary as standard correspondence where a material model exists, while preserving LCR's contribution as addressability and obligation bookkeeping.	Quantitative material behavior requires model parameters, units, citations, and data.
Continuum Sampling Bridge	This paper may bridge discrete and continuous language only as far as its sampling, projection, or boundary rule supports.	Sampling/interpolation rules, norms, error bounds, boundary conditions, units, and calibration routes.	Promote bridge claims only when sample rule, norm, and error or counterexample policy are stated.	Loose continuum, GR, plasma, or energy language remains presentation-level or calibration-bound.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
SUMMARY-II-Folded-Strand-Physics: Summary Paper II — Folded Strand Physics: The Gluon as Quark, Mass, Curvature, Tower, and Moonshine	75	This paper presents the physics applications of the Gluon formalism. We do not use the word "Gluon" because the fit is good. We use it because the substrate IS SU(3), and the physics IS the **standard model ga...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-16: Paper 16 - Continuum Edge Residuals	73	This paper defines continuum edge residuals as local window receipts. The closed result is local: every chart state anneals to a Lie-conjugate rest state in at most three S3 steps, the edge residue is exactly the st...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FORMAL: Paper 17 — C-Form: E6-E8 Error-Correction Tower Gluon	65	C = the tower Gluon — the Gluon that transports the error-correction state up the E6→E7→E8 tower. In the substrate, C is realized as the tower Gluon that: - Each tower level (E6, E7, E8) has its own Gluon `C_E...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-PAPER-016-C-Event-Enumeration s: CQECMLPX Unified Corpus — Paper 16 of 16	65	This capstone paper presents the complete catalogue of all predictive claims that follow from the single enumerated C-coordinate — the gluon invariant C that is preserved under all LCR readout operations. We show that...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.

Evidence card	Score	Publication contribution	Boundary
CQE-paper-16.25: Paper 16.25 - Toolkit for Continuum Edge Residuals	65	This support paper exposes the tools for Paper 16. The proof is the local edge-residual receipt; this toolkit shows how to inspect it. Run: ``text python production/formal-papers/CQE-paper-16/verify_continuum_edge_re...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	12 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series **Formalizing LCR, Unifying Standard Models**. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	FLCR-17
Legacy source slot(s)	16
Ribbon role	order-2 slot-6: bind state into causal, observer, or synchronization ledgers
Proof form	ledger, graph, and synchronization proof
Received state	state emitted by prior slot 15 and reopened at original slot 16 (Continuum Edge Residuals)
Produced state	formal result of original slot 16 plus the same-family lift path toward slot 26

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 17.1	normal_form_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	edge residuals can be tracked as finite bridge artifacts without asserting a global continuum limit
Proposition 17.2	normal_form_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	continuum-edge residual can be imported by later papers only through its declared source, receipt, and lane.
Protocol 17.3	falsifier_or_open_obligation	Receipt, source card, validator, citation, or CAM route named in the paper manifest	global continuum collapse and unbounded McKay/O2-prime closure remain separated residues

Figure Plan

Figure	Purpose	Caption
FLCR-17-F1	Causal/obligation ledger graph	Schematic showing how FLCR-17 turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-17-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-17-F3	Same-family lift context	Diagram placing this paper in the original 00-79 ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-17-F1: Causal/obligation ledger graph

```
received state
-> Causal/obligation ledger graph
-> formal claim lane
-> evidence object / receipt / citation
-> produced state
```

-> remaining residue

Table Plan

Table	Purpose
FLCR-17-T1	Obligation ledger table
FLCR-17-T2	Claim-lane/evidence-boundary table
FLCR-17-T3	Archive evidence card and source-backed upgrade table
FLCR-17-T4	Workbook replay and falsifier table

Worked Example

Convert one obligation into event, dependency, validator, and downstream route rows. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
causal edge	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
ledger	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
negative receipt	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
obligation	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the ledger_action proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-17-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?

Check	Reviewer question
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

FLCR-17 (Continuum Edge Residuals) occupies the ledger_action role at lift depth 1. The paper receives state emitted by prior slot 15 and reopened at original slot 16 (Continuum Edge Residuals). It produces formal result of original slot 16 plus the same-family lift path toward slot 26. The state transition is: bind state changes into causal, observer, synchronization, or obligation ledgers.

The paper-local proof form is:

ledger, graph, and synchronization proof

The integration result is a causal ledger that preserves dependency, synchronization, and obligation history. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
directed graphs and partial orders	Formal carrier for the paper-local result.
causal/event ledgers	Formal carrier for the paper-local result.
synchronization protocols	Formal carrier for the paper-local result.
traceability and provenance invariants	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 17.1	normal_form_result	edge residuals can be tracked as finite bridge artifacts without asserting a global continuum limit
Proposition 17.2	normal_form_result	continuum-edge residual can be imported by later papers only through its declared source, receipt, and lane.
Protocol 17.3	falsifier_or_open_obligation	global continuum collapse and unbounded McKay/O2-prime closure remain separated residues

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	16_Continuum_Edge_Residuals.md, virtuous/16_Continuum_Edge_Residuals_VIRTUOUS.md, supplements/16_INTERNAL_REASONING_SUPPLEMENT.md	routed evidence

Evidence class	Routed items	Status
Evidence inputs	CAM_CLAIM_100_SCORE_AUDIT.md, NSIT_TOOL_CLOSURE_MATRIX.md, NSIT_REASONED_CLOSURE_PASS.md, PAPER_REWORKS_CRYSTAL_PROJECTION.json	routed evidence
CAM/crystal sources	PAPER_REWORKS_CRYSTAL_PROJECTION.json, AGENT_CRYSTAL_WORK_HARVEST.json, runtime standards artifact, notebook-derived source bundle	routed evidence
Standards-window inputs	window_summary.json, node DBs, window DBs	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting the operation;
4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-17-F1: Causal/obligation ledger graph	Visualizes the proof object, routing, or boundary.
FLCR-17-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-17-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-17-T1: Obligation ledger table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-17-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-17-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-17-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.